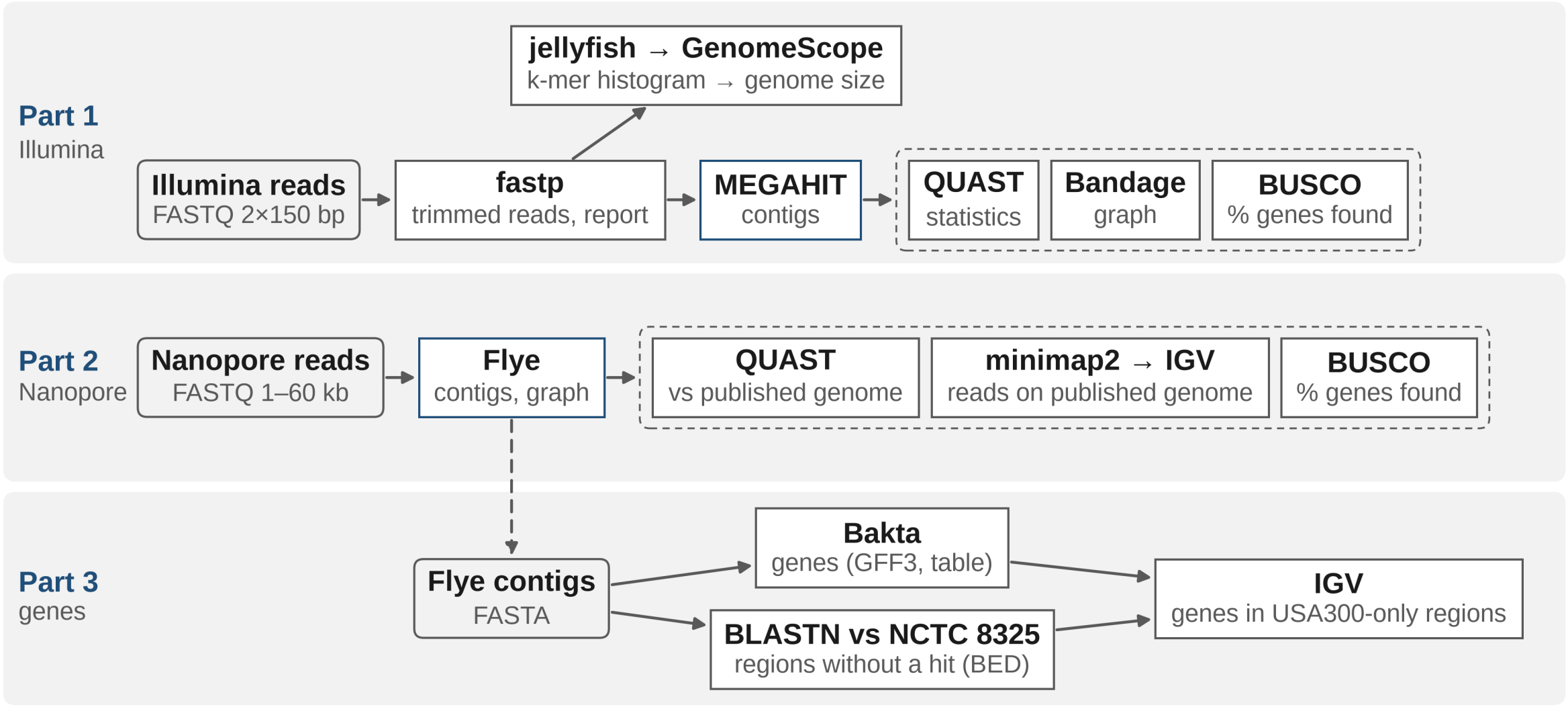
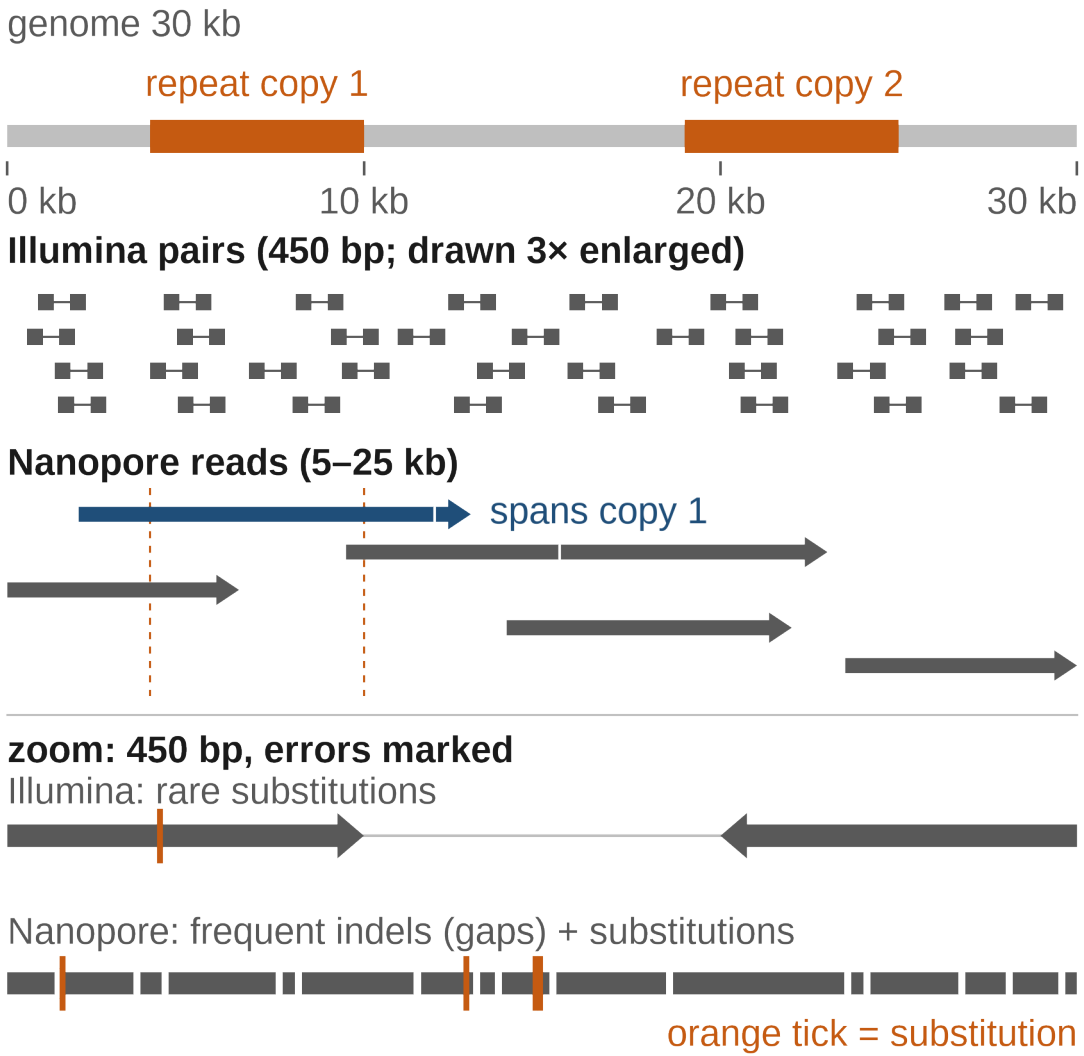
The data and the pipeline

From reads to an annotated genome



Two kinds of reads, one isolate

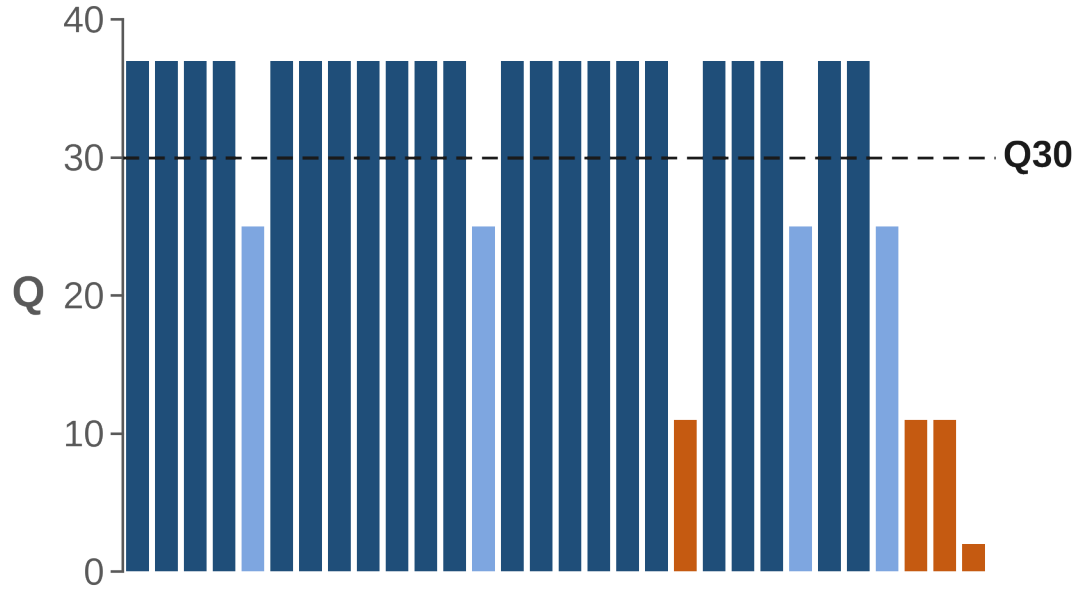
- **Illumina** 2×150 bp pairs, error < 0.5 %, mostly substitutions
- **ONT** reads of 1–60 kb, error 2–5 %, mostly indels
- Our isolate may differ from the published genome



Part 1 – Illumina reads

A FASTQ record and its qualities

	instrument	flow cell	lane	tile	x : y	read 1
read name	@A00917	:12:HKV7TDSX7	:1	:1101	:10746:1016	1:N:0:GATCAG
sequence	TAAAATTGAACCGCCAGGAACCTGCCGGAG					
separator	+					
qualities	FFFF	:FFFFFFF	:	FFFFFF	,FFF:FF:,,#	



Phred+33: p = 10^(−Q/10)

char	ASCII	− 33 = Q	error prob.	1 error in
F	70	37	0.02 %	5 000
?	63	30	0.1 %	1 000
:	58	25	0.32 %	320
,	44	11	7.9 %	13
#	35	2	63 %	1.6

Adapters appear when the insert is short

insert 450 bp

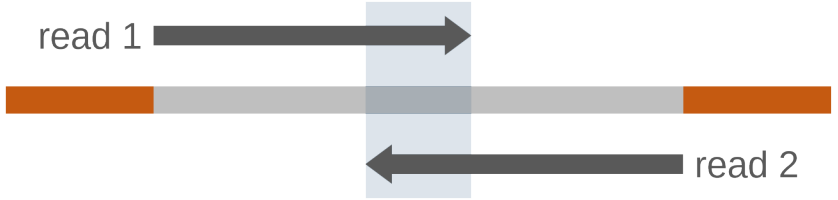


reads do not overlap

fastp: “not overlapped”

insert size unknown

insert 250 bp



reads overlap by 50 bp

insert = 150 + 150 − 50
= 250 bp

insert 100 bp



insert < read length

each read ends with
50 bp of adapter → trim

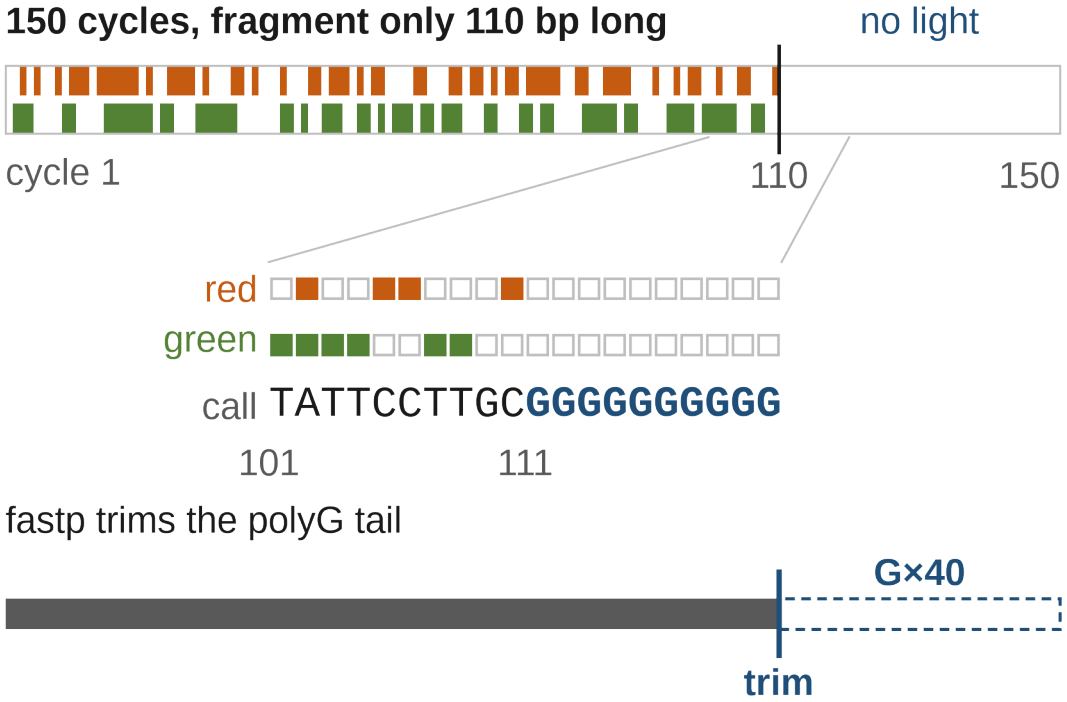
end of read 1: GACACTGAGATTAGATCGGAAGAGCACACG…
AGATCGGAAGAGC… = TruSeq adapter

polyG: a dark cluster is read as G

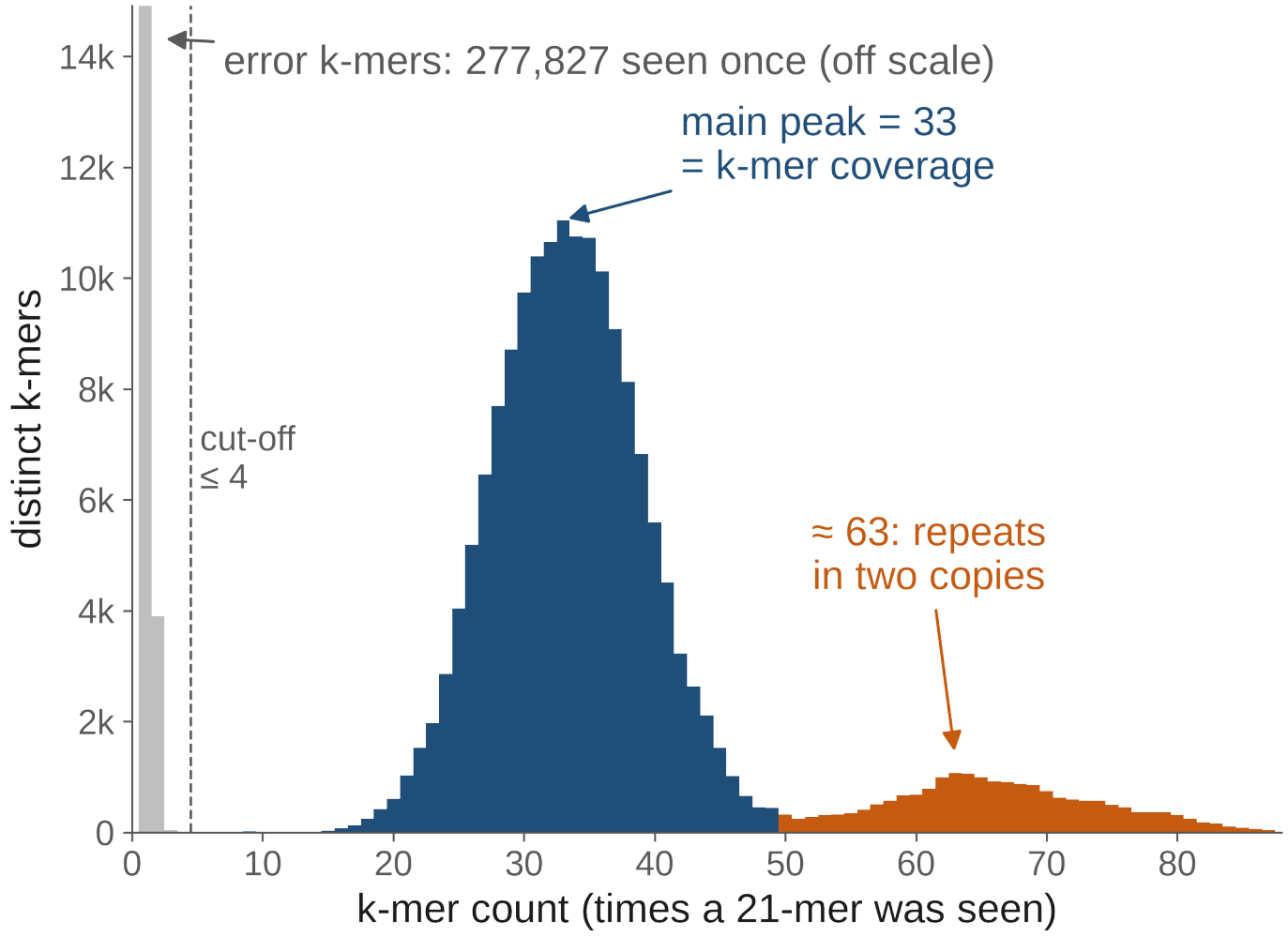
- NovaSeq uses **two colours** for four bases
- G = no light in either channel
- Fragment ended → no light → GGGGG…
- fastp trims polyG tails of NovaSeq reads automatically

base	red	green
A	■	■
C	■	□
T	□	■
G	□	□

G = no light



Reading a k-mer spectrum



Genome size from the spectrum

all k-mers (Σ count × number)	6,933,290
− error k-mers (count ≤ 4)	285,742
= k-mers from the genome	6,647,548
÷ main peak	33
≈ genome size	**201 kb**

(toy genome: 200 kb, haploid)

k-mer coverage < read coverage

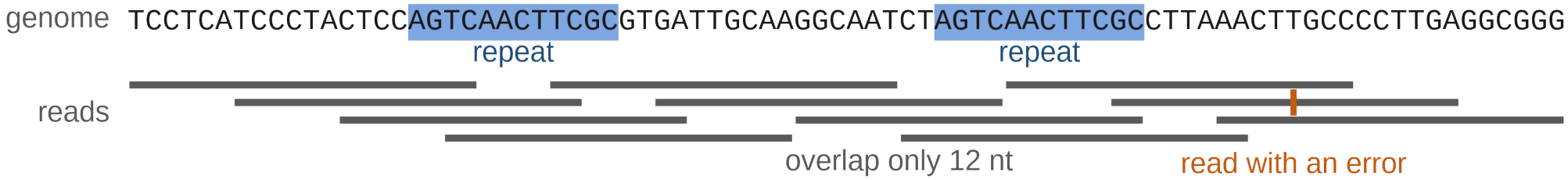
a read of L bases holds L − k + 1 k-mers:

k-mer cov. = read cov. × (L − k + 1) / L

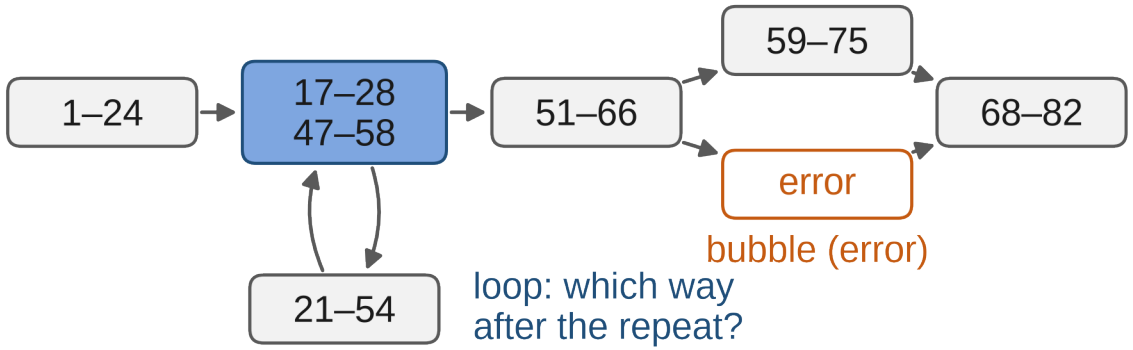
= 40× × 130/150 = 34.7×

k-mers hit by errors lower it to 33×

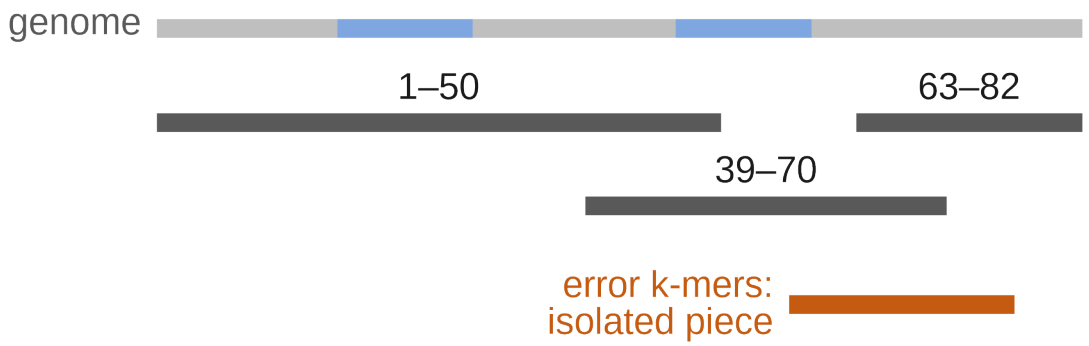
k in a de Bruijn graph: small or large?



small k = 9: repeat collapses, graph branches



large k = 15: repeat resolved, graph breaks



3 contigs, no links: thin overlap and the error remove the 15-mers that joined them

MEGAHIT: k=21 → k=29 → k=39 → k=59 → k=79 → k=99 → k=119 → k=141

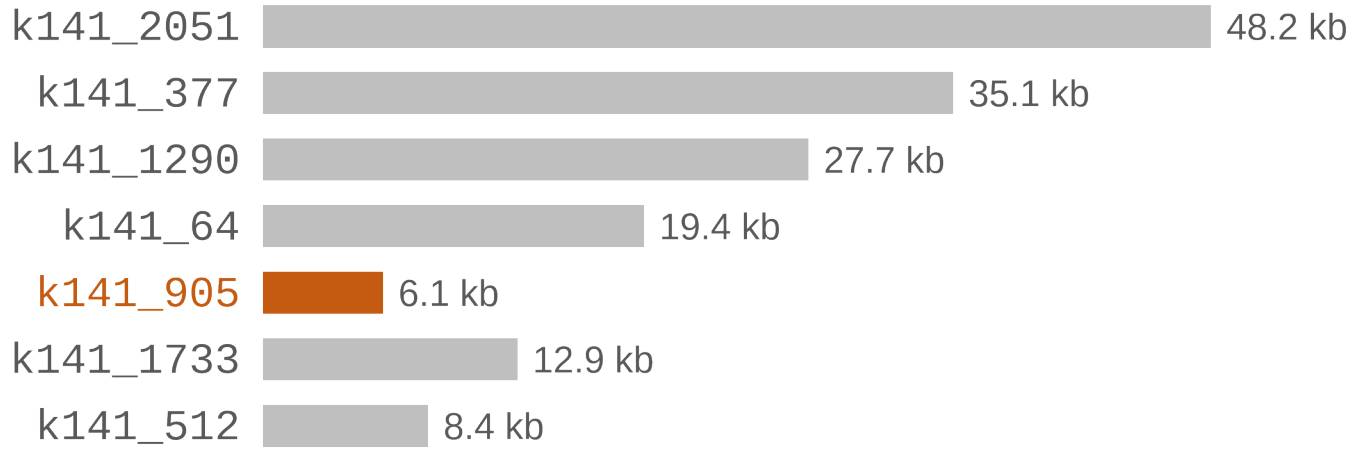
contigs of each round are input to the next

MEGAHIT contig names carry coverage

```
>k141_2051 flag=0 multi=12.3071 len=48211
```

- k-mer size of the last round
- contig id
- 0 = joined to other contigs; 1 = stand-alone; 3 = circular (loop)
- average k-mer coverage (multiplicity) of the contig
- length in bp

toy assembly (one row = one contig)

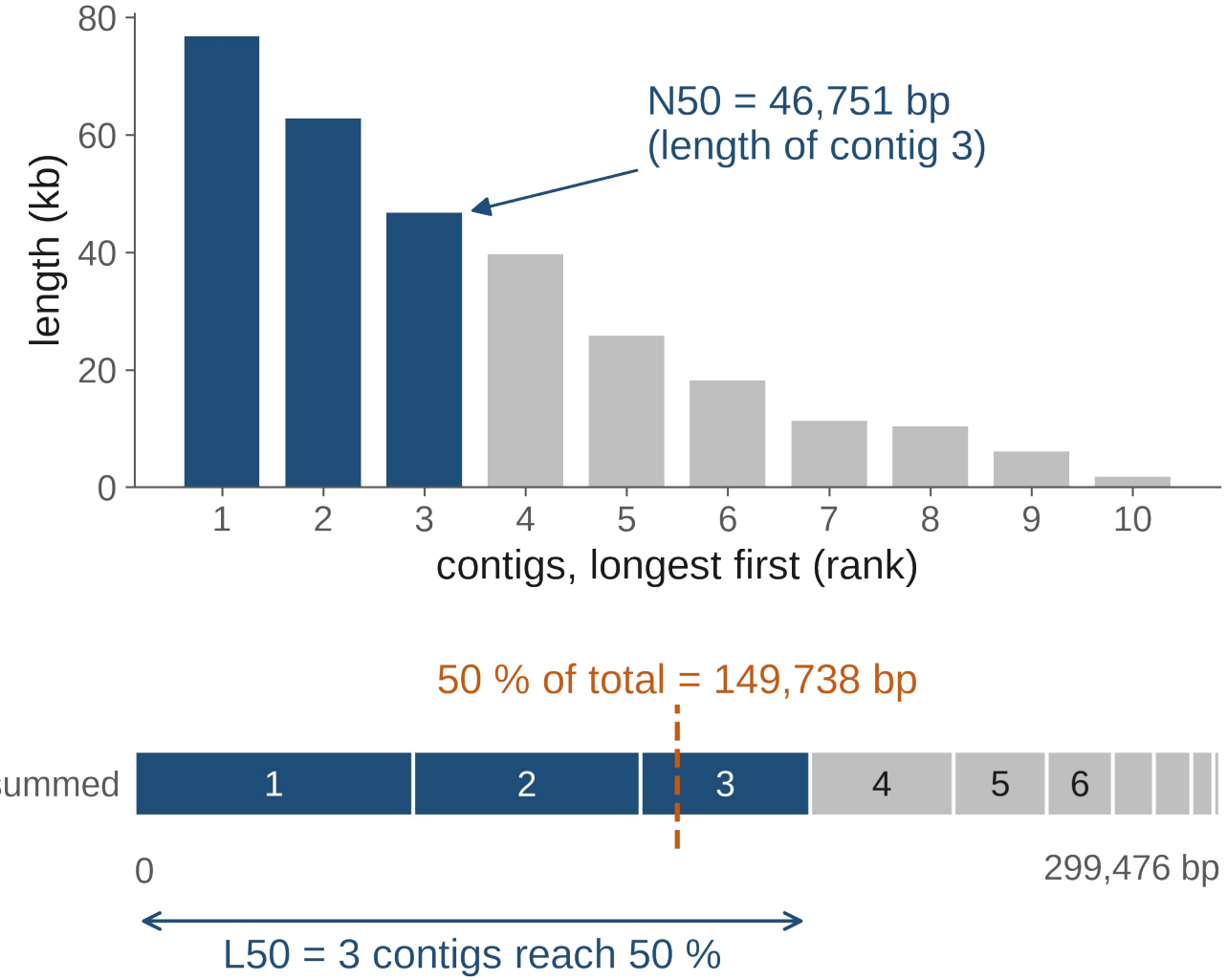


	multi
k141_2051	12.3
k141_377	12.0
k141_1290	12.7
k141_64	11.4
k141_905	**24.3**
k141_1733	11.8
k141_512	12.4

≈ 12 → single copy

≈ 2 × 12 → sequence present twice in genome

N50 and the QUAST report

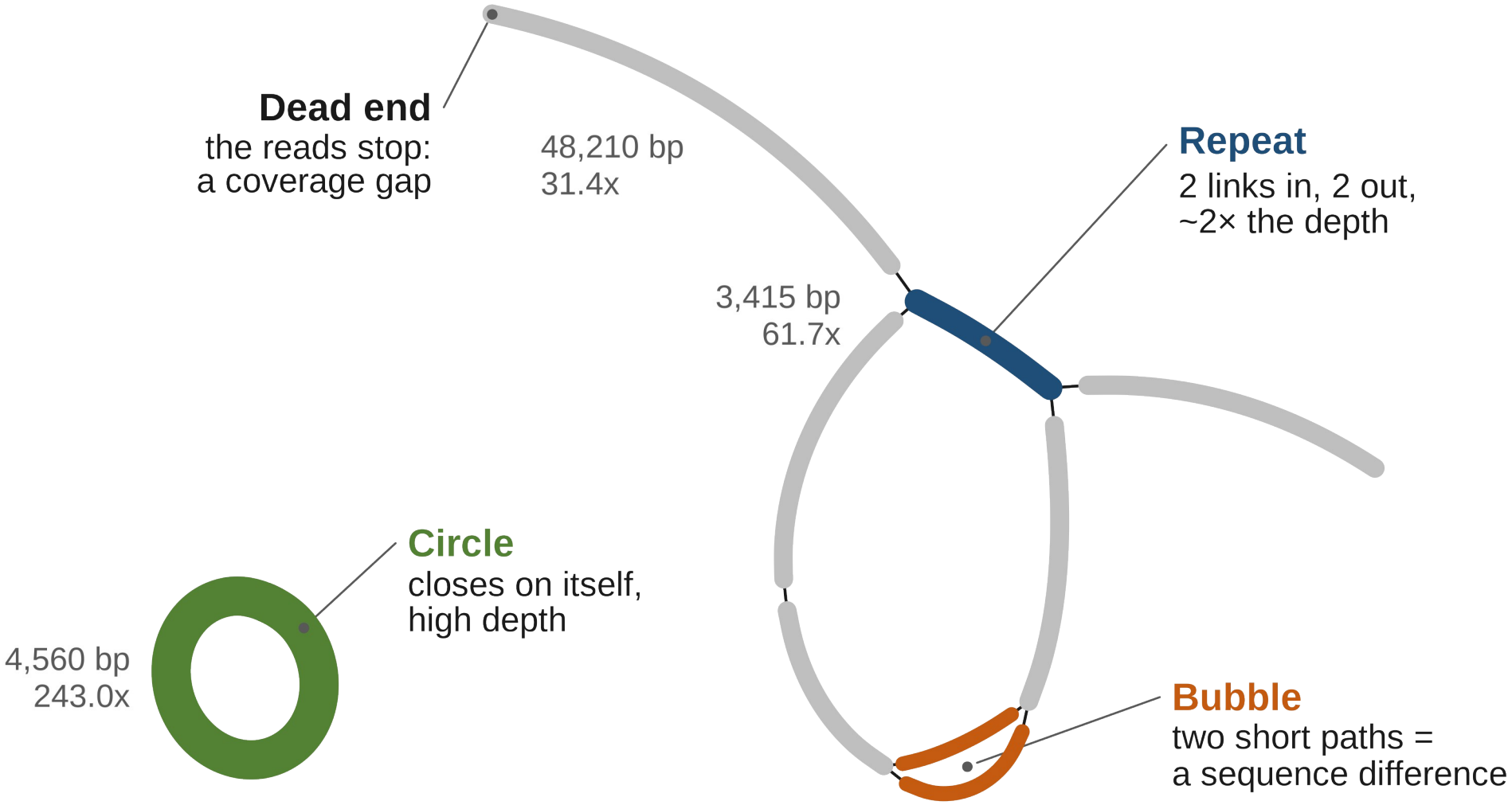


QUAST report.txt

Assembly	contigs
# contigs (>= 0 bp)	10
…	
# contigs	10
Largest contig	76840
Total length	299476
…	
N50	**46751**
…	
L50	**3**

contigs < 500 bp are ignored, except in the "(>= 0 bp)" lines

The assembly graph in Bandage



Part 2 – Nanopore reads

Flye's assembly_info.txt

#seq_name	length	cov.	circ.	repeat	mult.	alt_group	graph_path
contig_1	4120550	**38**	**Y**	N	**1**	*	1
contig_2	31870	**76**	**Y**	Y	**2**	*	2
contig_3	18240	**37**	**N**	N	**1**	*	3

length in bp

mean read coverage

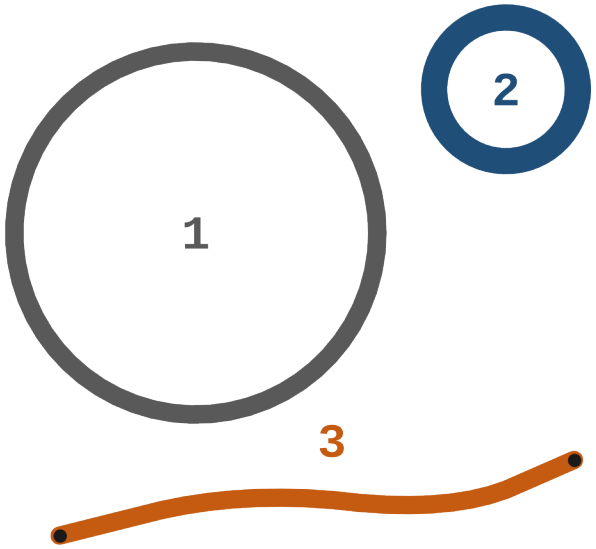
Y = path closes a loop

Y = repeat in the graph

copies, from coverage

* = primary contig

edge numbers in the graph



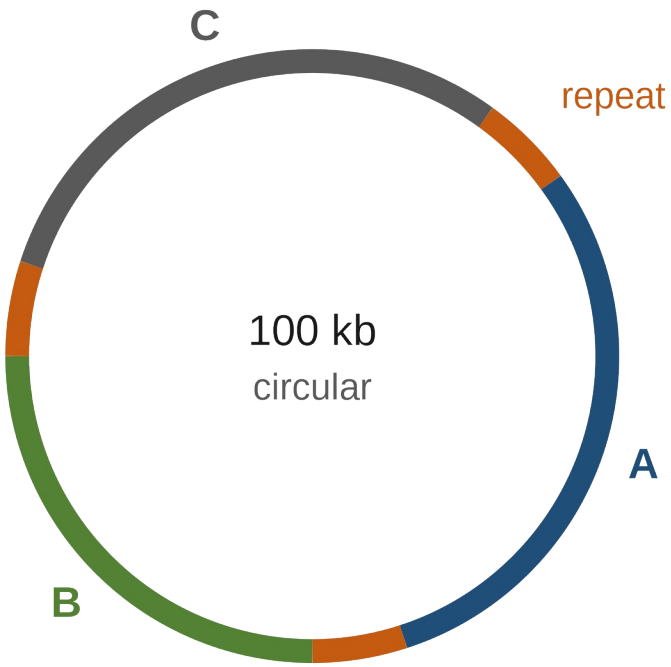
linear: two dead ends

`cov. 76 / 38 = 2`

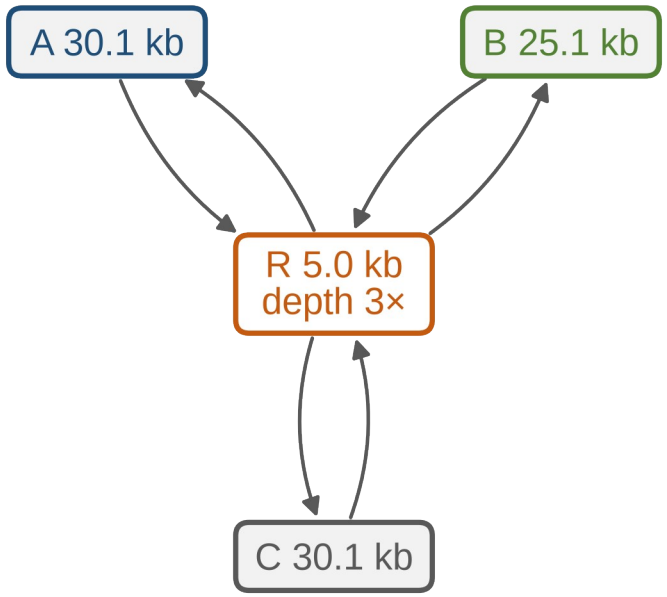
contig_2: 2 copies per chromosome

Why long reads give complete genomes

Genome: 3 copies of a 5 kb repeat

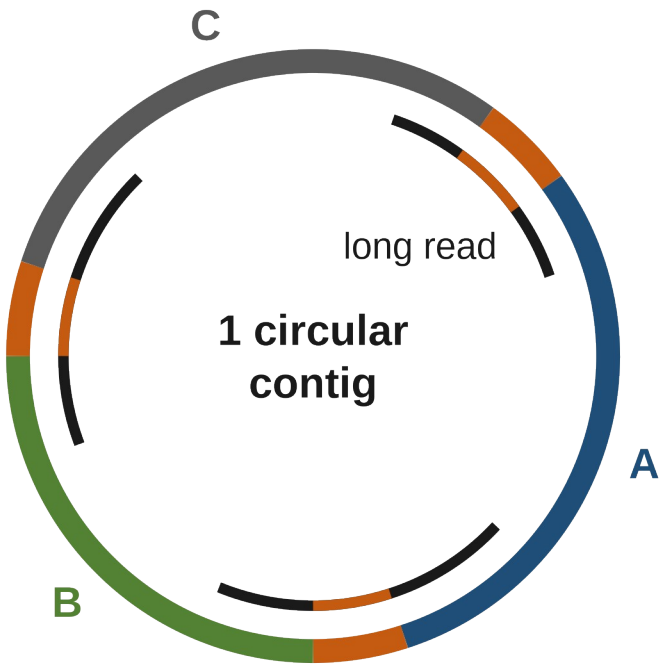


Short reads (k = 31): graph



3 ways into R, 3 ways out:
4 contigs (A, B, C and R)

Long reads (6–20 kb)



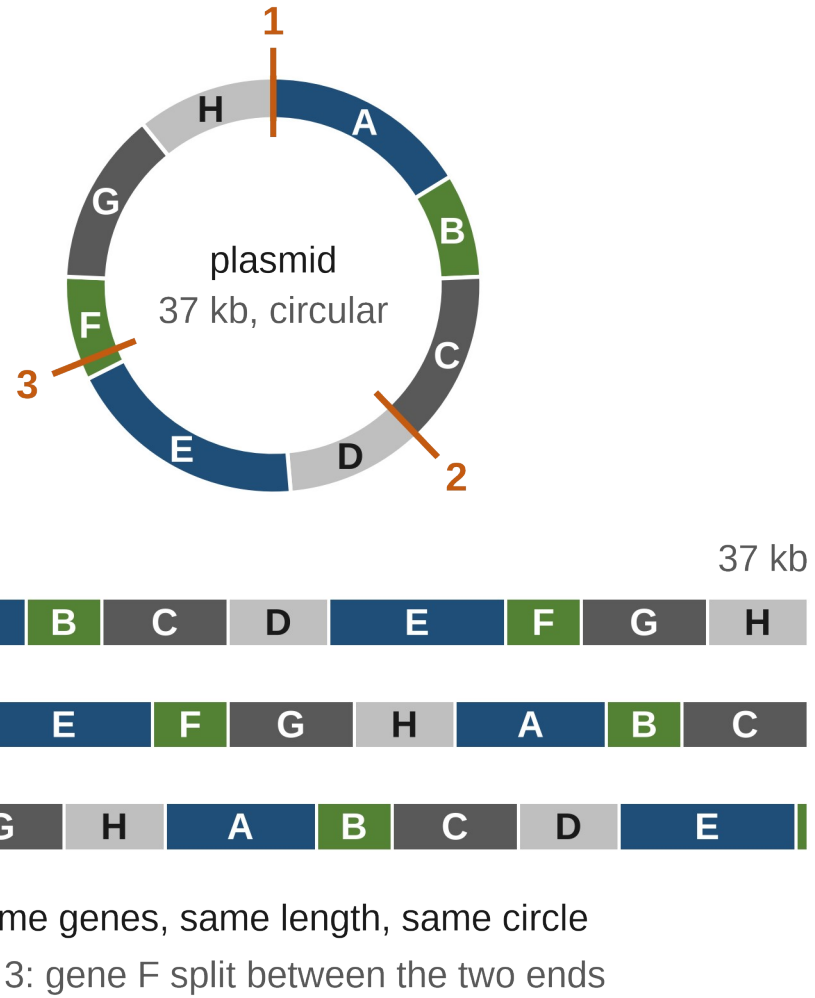
each read crosses a repeat copy
with unique ends: A → R → B → R → C → R → A

Same data – same result?

- Programs running on several threads are often **not deterministic**
- Threads finish in a different order → slightly different output
- **May differ**: contig names, start of a circular contig, rounded coverage
- **Must not differ**: sequence, contig lengths, genes found
- Your result may differ from your neighbour's or the teacher's

one plasmid, three runs

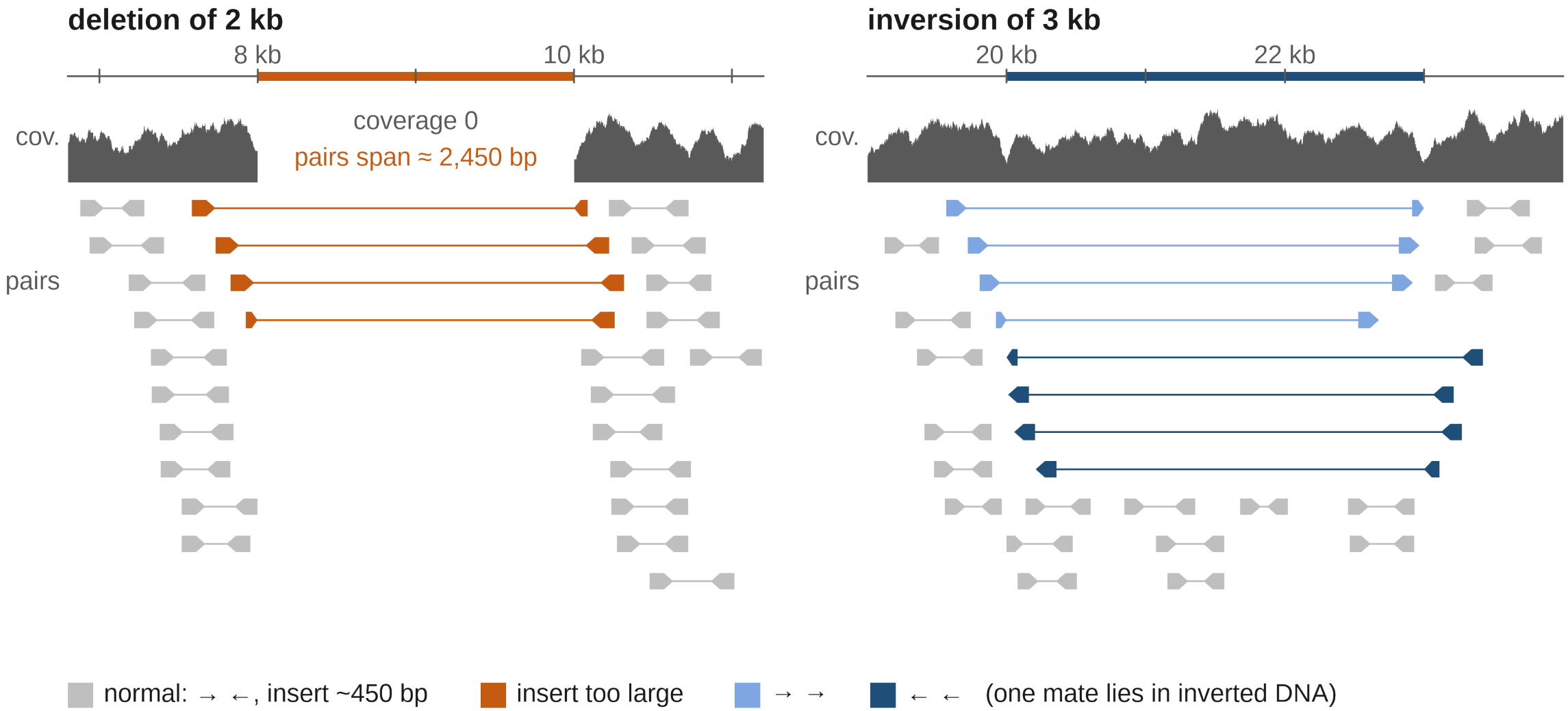
cut = start of contig



same genes, same length, same circle

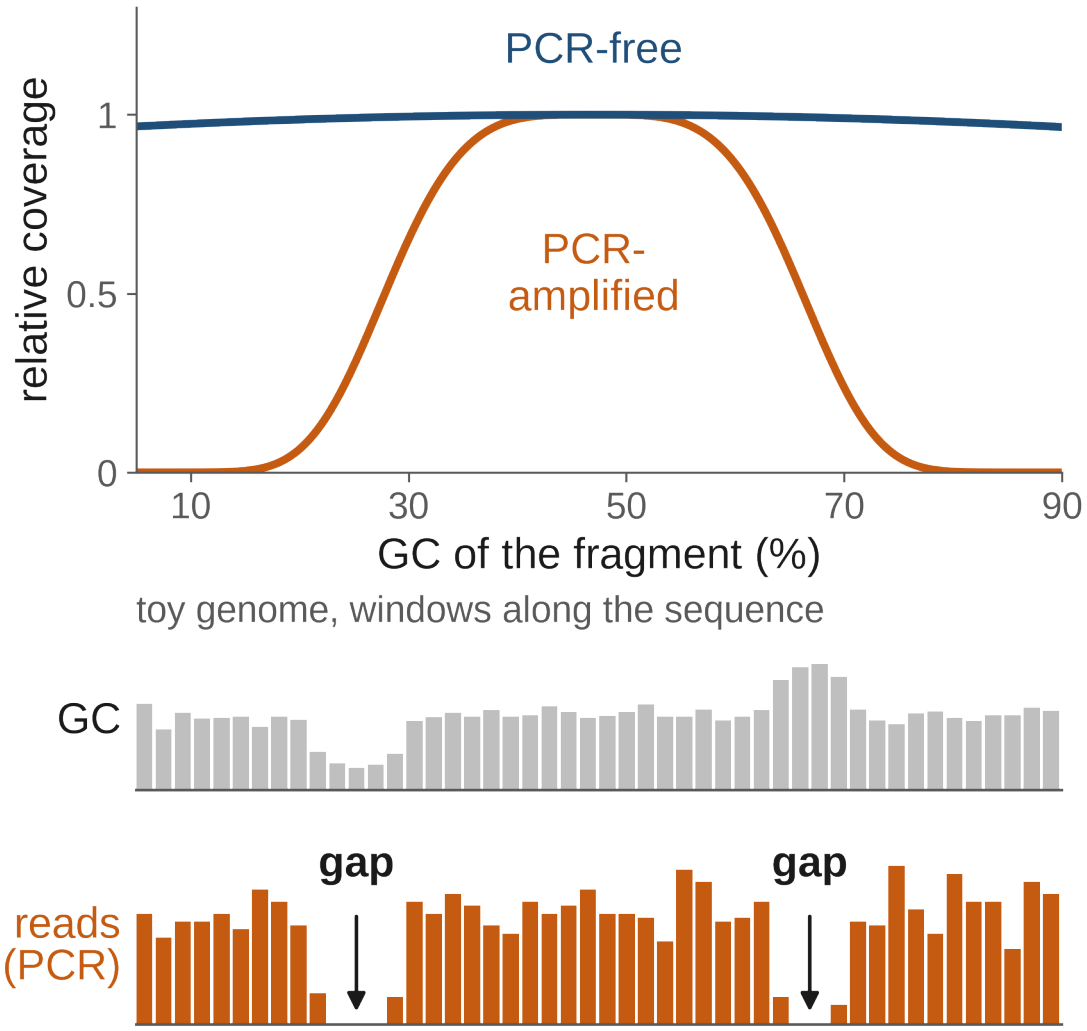
run 3: gene F split between the two ends

Read pairs in IGV: normal and unexpected



Why coverage is not even

- Illumina libraries are amplified by **PCR**
- Fragments with extreme GC amplify poorly
- Low coverage → missing k-mers → gaps
- PCR-free libraries (most ONT kits) are even



Aird et al. 2011, Genome Biol 12:R18

BUSCO: complete, fragmented, missing

1 Lineage set

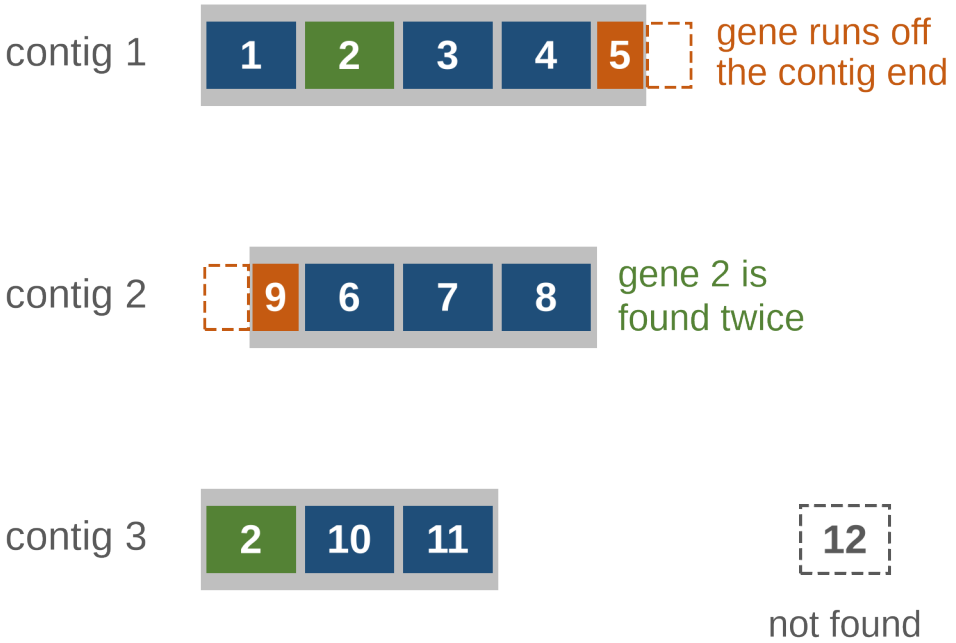
genes found once in nearly every genome of the group

staphylococcus_odb12: 1317 genes

toy set: 12 genes

1	2	3	4
5	6	7	8
9	10	11	12

2 Search the assembly



3 Classify and count

S	complete, single-copy	8
D	complete, duplicated	1
F	fragmented	2
M	missing	1
n	genes searched	12

summary line:

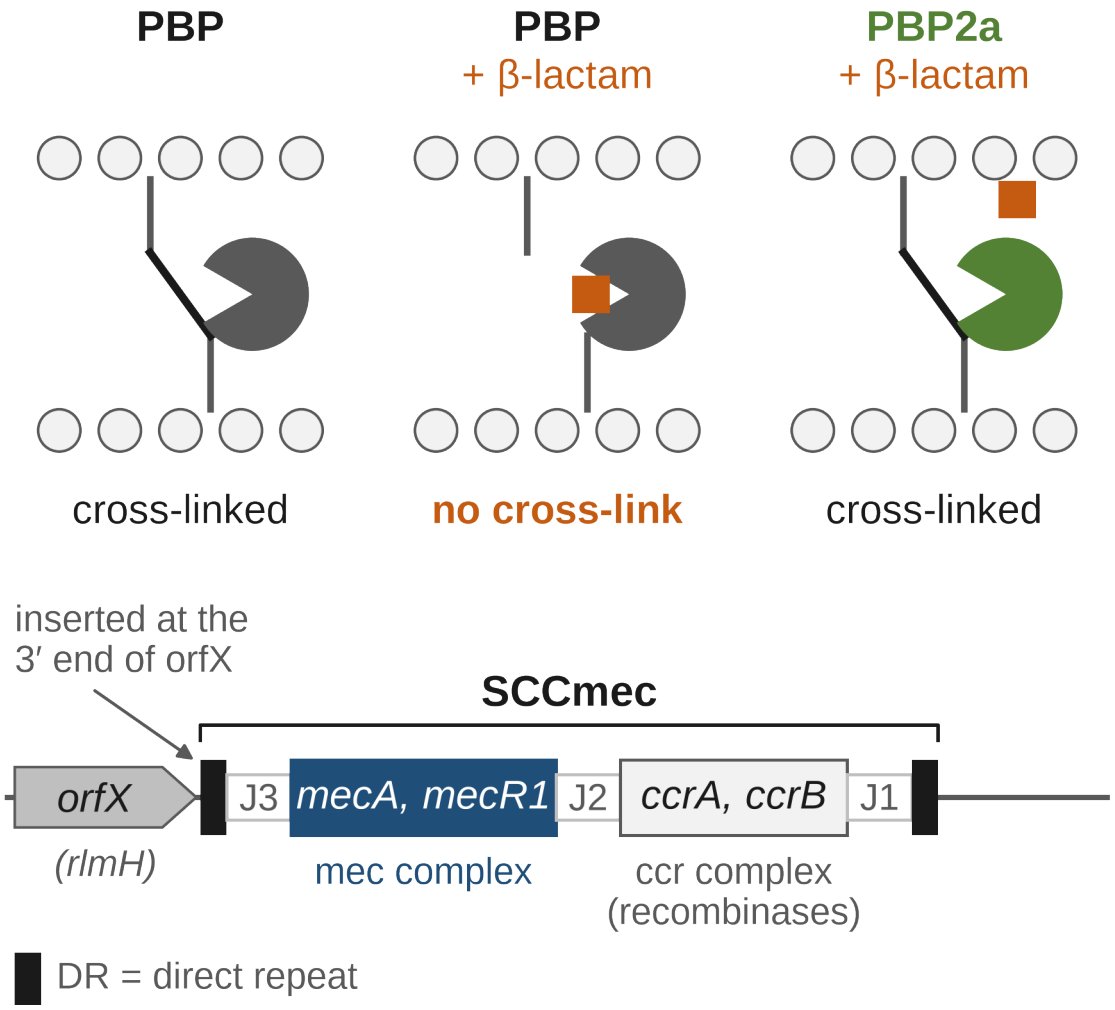
`C:75.0%[S:66.7%,D:8.3%], F:16.7%,M:8.3%,n:12`

C = S + D = 9/12

Part 3 – MRSA

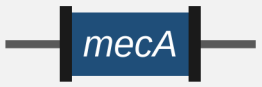
How methicillin resistance works

- β-lactams block the **PBPs** that cross-link the cell wall
- **PBP2a** (gene *mecA*) binds β-lactams poorly
- *mecA* sits on a mobile element: **SCCmec**
- SCCmec integrates at one site of the chromosome



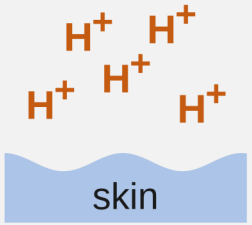
USA300: a community MRSA

- Most common community MRSA in North America
- Skin and soft-tissue infections in healthy people
- Known for mobile elements: SCCmec IV, ACME, PVL phage
- Compared with lab strain NCTC 8325 (methicillin-sensitive)



SCCmec IV

mecA → PBP2a:
resistance to β-lactams



ACME

arginine catabolic mobile element:
survives acidic conditions,
tolerates skin polyamines

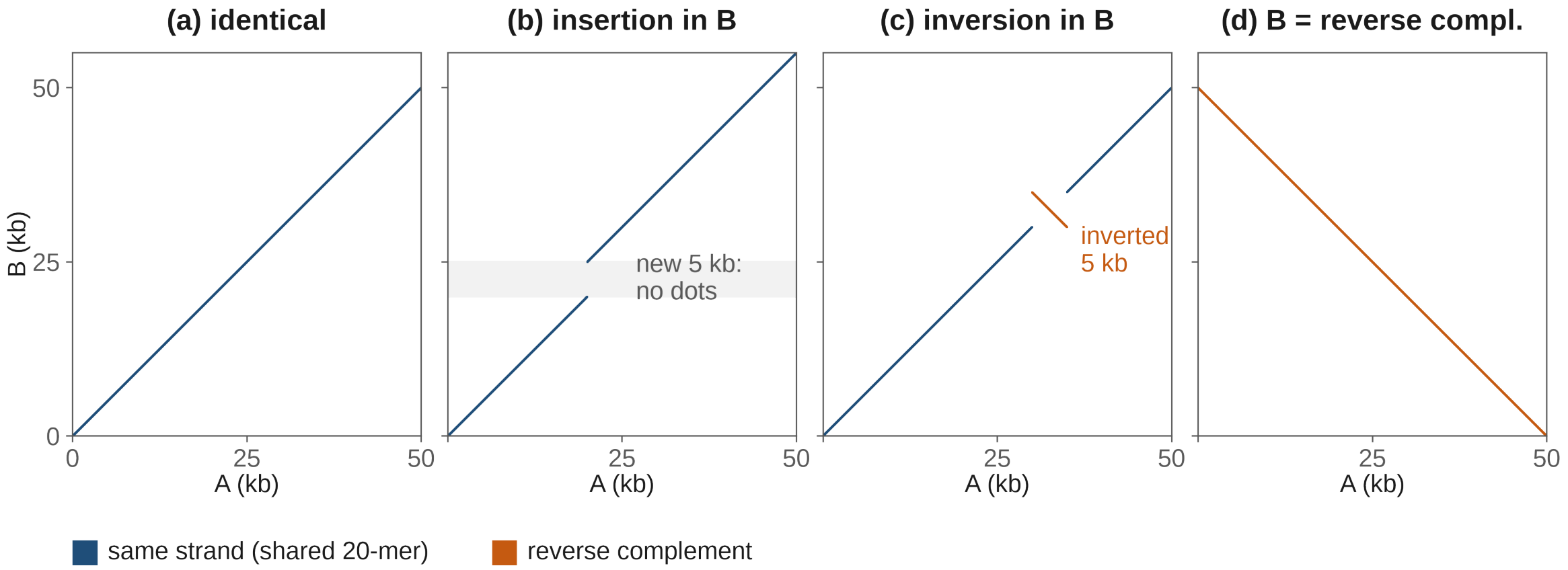


PVL prophage

Panton–Valentine leukocidin:
toxin that kills white blood
cells (neutrophils)

Planet et al. 2013, mBio 4:e00889-13; Glaser et al. 2016, mBio 7:e02183-15

Reading a genome dotplot



BED files: intervals on a genome

- **BED** = one interval per line: `contig start end` (tab-separated, more columns optional)
- Start is **0-based**, end is not included
- BLAST, GFF and IGV count from **1** and include both ends
- Length = end − start
- IGV, bedtools and seqkit read BED files directly

one feature, two ways to write it



BLAST, GFF, IGV **4 – 7**

first and last base

BED **3 7**

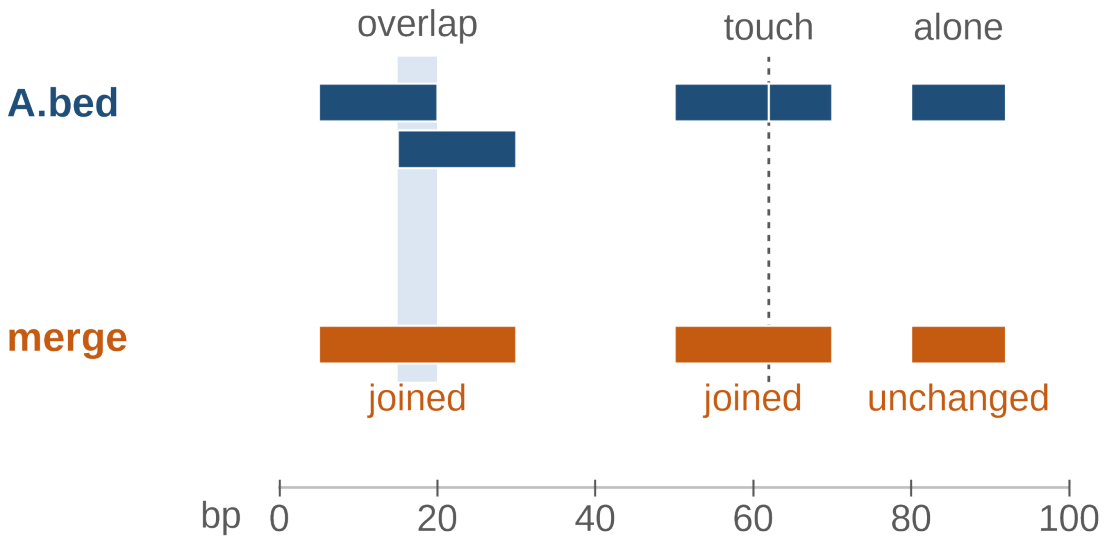
start = first − 1, end = last

length = end − start = 7 − 3 = 4 bp

UCSC FAQ "BED format"; bedtools documentation (bedtools.readthedocs.io)

bedtools merge: join overlapping intervals

- `bedtools merge -i A.bed`
- Overlapping intervals become **one**
- Touching intervals (end = next start) too
- Input must be **sorted** by contig and start
- Used in step 3.3: many BLAST hits → shared regions



A.bed

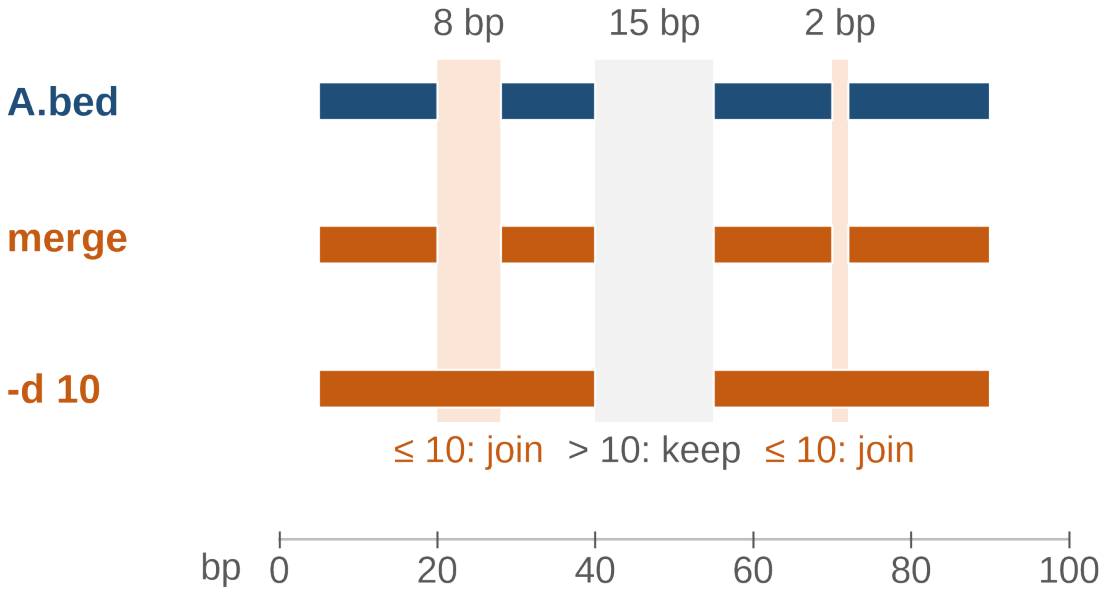
```
chr1    5   20
chr1   15   30
chr1   50   62
chr1   62   70
chr1   80   92
```

merge result

```
chr1    5   30
chr1   50   70
chr1   80   92
```

merge -d: bridge small gaps

- `bedtools merge -i A.bed -d 10`
- Intervals at most **d bp apart** are joined too
- Larger gaps stay
- Step 3.3 uses `-d 1000`: gaps under 1 kb disappear, so only regions over 1 kb remain



A.bed

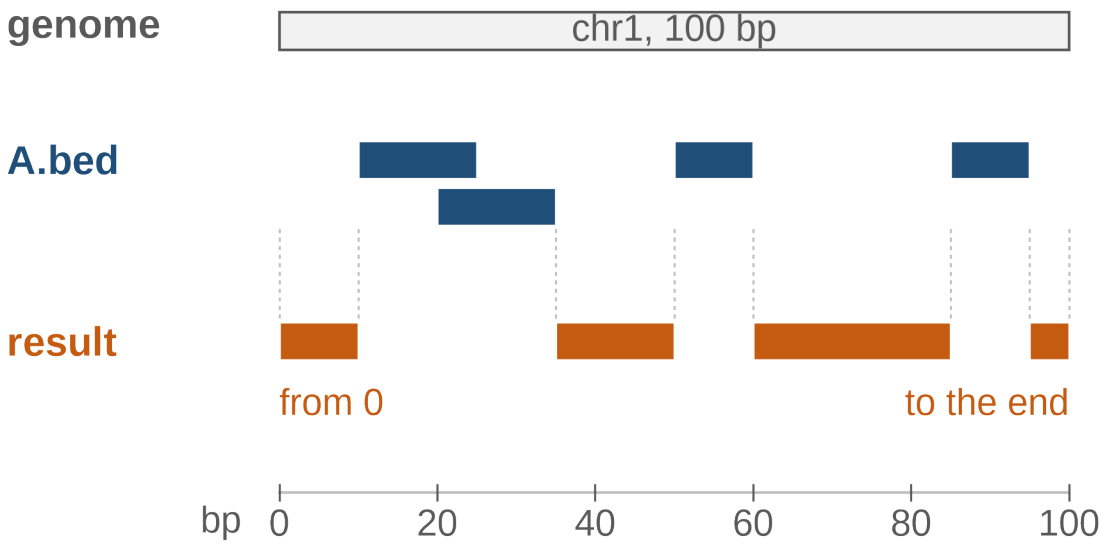
```
chr1    5   20
chr1   28   40
chr1   55   70
chr1   72   90
```

merge -d 10

```
chr1    5   40
chr1   55   90
```

bedtools complement: what is not covered

- `bedtools complement -i A.bed -g genome.txt`
- Everything **not** in A
- Needs a **genome file**: contig, length
- Includes the contig start and end
- Step 3.3: not shared = USA300-specific



genome.txt

```
chr1   100
```

A.bed

```
chr1    10    25
chr1    20    35
chr1    50    60
chr1    85    95
```

result

```
chr1     0    10
chr1    35    50
chr1    60    85
chr1    95   100
```

bedtools documentation, complement (bedtools.readthedocs.io)

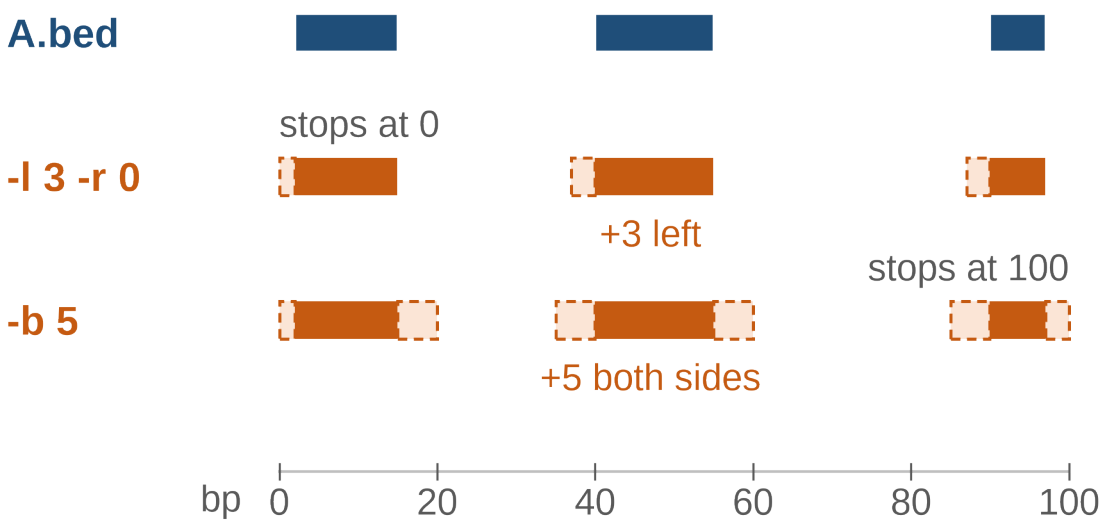
bedtools slop: make intervals longer

- `bedtools slop -i A.bed -g genome.txt -l 3 -r 0`

- Adds bases to the left (`-l`), right (`-r`) or both sides (`-b`)

- Stops at the contig start and end (genome file)

- Step 3.3: `-l 1 -r 0` turns 1-based BLAST starts into BED starts



A.bed

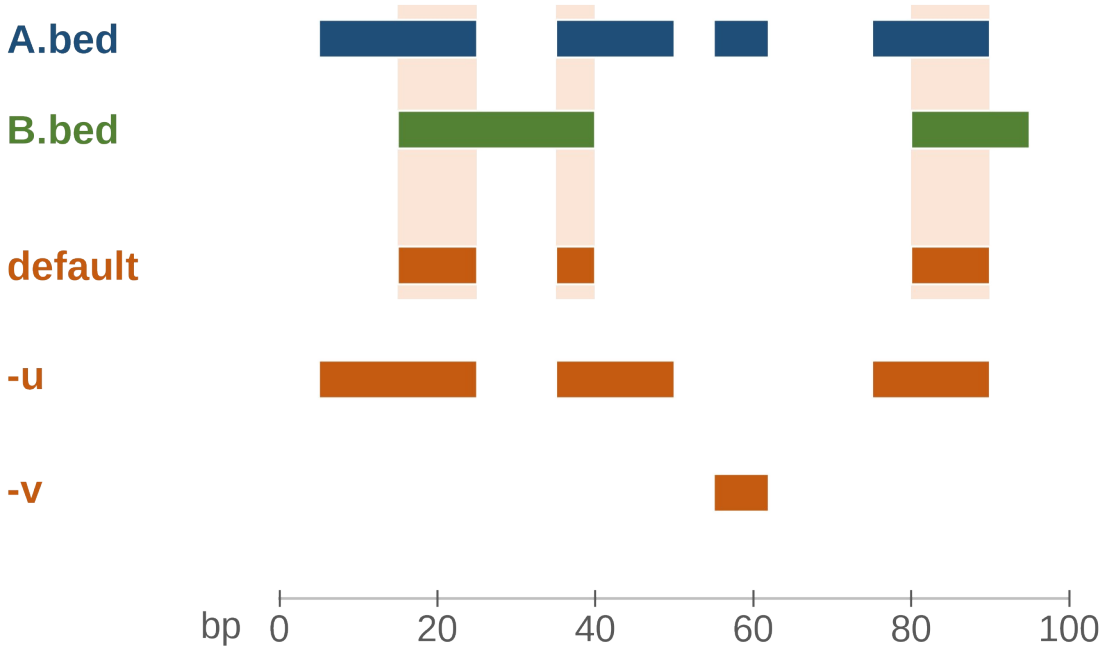
```
chr1    2    15
chr1   40    55
chr1   90    97
```

slop -l 3 -r 0

```
chr1    0    15
chr1   37    55
chr1   87    97
```

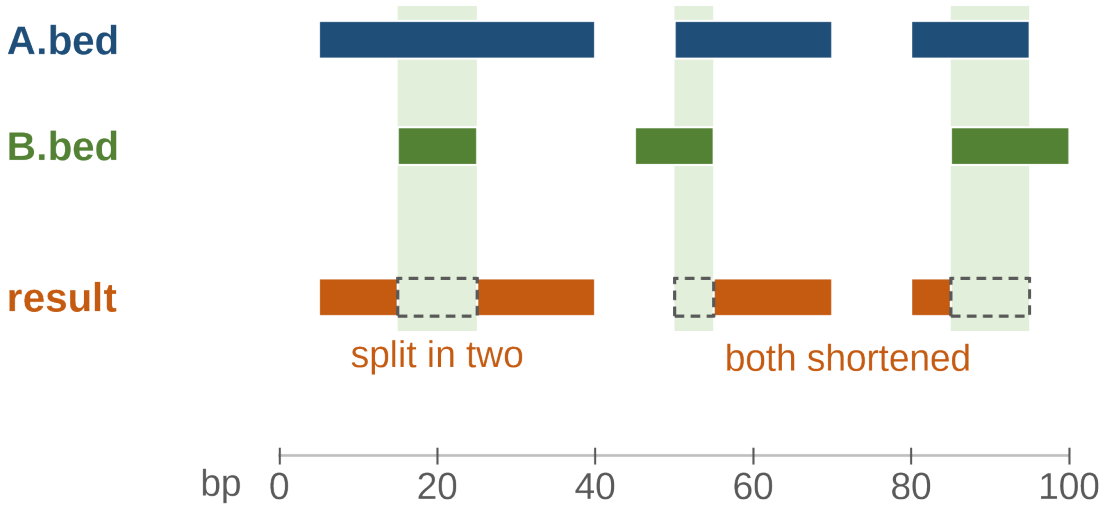
bedtools intersect: where A and B overlap

- `bedtools intersect -a A.bed -b B.bed`
- Default: only the **overlapping parts**
- `-u`: whole A intervals that overlap B
- `-v`: A intervals that do **not** overlap B
- Typical use: genes inside regions (`-a genes -b regions -u`)



bedtools subtract: remove what B covers

- `bedtools subtract -a A.bed -b B.bed`
- Cuts away every part of A covered by B
- An interval can be **shortened** or **split** in two
- Intervals without overlap stay



A.bed			B.bed			result		
chr1	5	40	chr1	15	25	chr1	5	15
chr1	50	70	chr1	45	55	chr1	25	40
chr1	80	95	chr1	85	100	chr1	55	70
						chr1	80	85

Step 3.3: from BLAST hits to BED regions

BLAST output: by score, 1-based

```
blastn ... -outfmt "6 qseqid qstart qend"
```

sort by contig and start

```
sort -k1,1 -k2,2n
```

start − 1: BED coordinates

```
bedtools slop -g genome.txt -l 1 -r 0
```

join: shared regions

```
bedtools merge -d 1000 > shared.bed
```

the rest: USA300-specific

```
bedtools complement -g genome.txt
```

first lines of the file

```
contig_1  40001   52000
contig_1  30001   41000
contig_1      1    8000
...
```

```
contig_1      1    8000
contig_1   7001   15000
contig_1  15401   22000
...
```

```
contig_1      0    8000
contig_1   7000   15000
contig_1  15400   22000
...
```

```
contig_1      0   22000
contig_1  30000   52000
```

```
contig_1  22000   30000
contig_1  52000   60000
```

